

Coordinated multiple *wh*-sluicing in English: A corpus-based investigation

Hee-Yeon Kim and Jong-Bok Kim

heeyeonkim@khu.ac.kr & jongbok@khu.ac.kr

(Kyung Hee University)

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Coordinated multiple *wh*-sluicing (CM-sluicing)

- English coordinated multiple sluicing phenomenon (CM-sluicing), is a type of sluicing in which two fragmental expressions are coordinated (Hankamer and Sag 1976, Ginzburg and Sag 2000, among others).
 - (1) Someone saw something, but I can't remember **who** and **what**. (Citko and Gračanin-Yuksek 2020: (5))
- It is often compared with other multiple *wh*-constructions such as coordinated multiple *wh*-question (CWHQ) that is quite restricted in English (Citko and Gračanin-Yuksek 2020).
 - (2) ***Who** and **what** saw? (Citko and Gračanin-Yuksek 2020: (2))

- CM-sluicing is not subject to clause-mate condition which requires all *wh*-remnants to originate in the same finite clause (Merchant 2013).
 - (3) a. [Something was served at the fair [because someone told us to]] but Lee doesn't recall **what** and **who**.
b. [[Someone served something at the fair]] but Lee doesn't recall **what** and **who**. (Adapted from Citko and Gračanin-Yuksek 2020: 13)
- This indicates that CM-sluicing is able to accommodate a more diverse set of structure as its antecedent.

SWIPING and CM-sluicing

- CM-sluicing has no restriction regarding SWIPING, which is an acronym for ‘**Sluiced Wh-word Inversion with Preposition in Northern Germanic**’ and is a distinct characteristic of sluicing (Citko and Gračanin-Yuksek 2020).
 - (4) I know John was talking with somebody about something,
 - a. ...but I don’t know **who with** or **about what**.
 - b. ...but I don’t know **with who** or **what about**.
- This is not applied to CWHQ, for it is a question, not an elliptical construction.
 - (5) a. ? * **Who with** or **about what** was John talking?
b. ? * **With who** or **what about** was John talking?
- This affirms that CM-sluicing is a type of sluicing and an elliptical construction, without restrictions to the p-stranding or pied-piping of the prepositions.

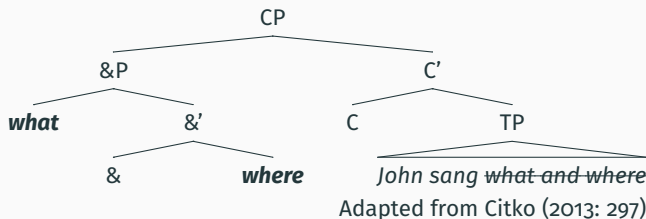
Restriction on *it* reading

- CM-sluicing can accommodate the *it* reading, but not the *at-all* reading, which implies that the sluicing process of the two remnants are done respectively (Citko and Gračanin-Yuksek 2020).
 - (6) Citko and Gračanin-Yuksek (2020: 7)
 - a. I heard that John sang something, but I forgot **what and where**.
 - b. # ... **what** he sang and **where** he sang **at all**. *at-all reading*
 - c. ... **what**he sang and **where** he sang **it/the thing he sang**. *it reading*
- Unlike CM-sluicing, both of these readings are possible in CWHQs, having the *at-all* reading implies that is possible for the two *wh*-phrases to be processed at the same time.
 - (7) Citko and Gračanin-Yuksek (2020: 7)
 - a. **What** and **where** did John sing?
 - b. **What** did John sing and **where** did John sing **at all**? *at-all reading*
 - c. **What** did John sing and **where** did John sing **it/the thing he sang**? *it reading*

Mono-clausal structure

- There can be three structures of CM-sluicing, because there are three of CWHQ (Citko and Gračanin-Yuksek 2020).
- First is the mono-clausal structure, in which the remnants form a complex &P and their putative source clause is processed all at once (Zhang 2007).

(8)

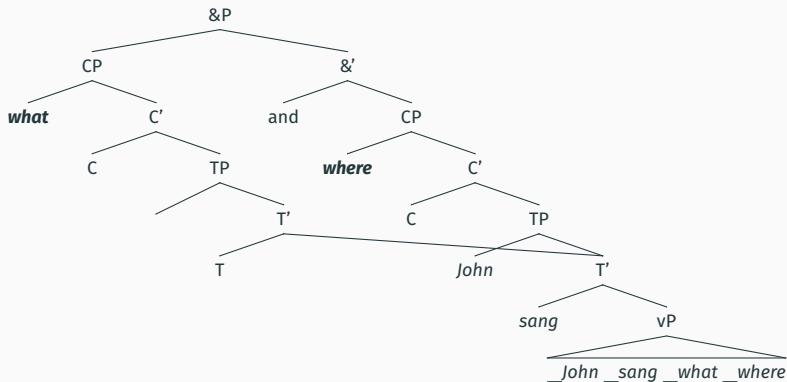


- It is impossible to account for the internal structures of the shared elements as it requires both of the clauses to be identical.

Bi-clausal bulk-sharing structure

- The remnants are realized into two different CPs, in which the sluiced elements in bulk are shared by the first remnant, having dominance over the second remnant (Citko 2013).

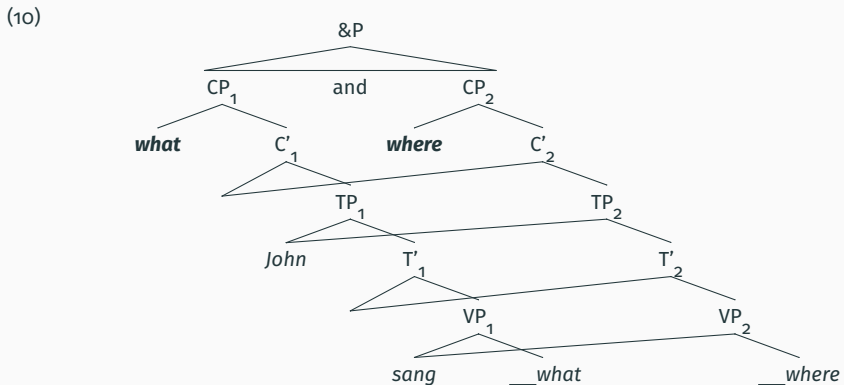
(9)



Adapted from Citko (2013: 310)

Bi-clausal non-bulk-sharing structure

- This structure shows multi-dominance between the two *wh*-phrases on each other, so the structure of the first remnant does not pose dominance on the second remnant (Citko 2013).



Adapted from Citko (2013: 309)

Research questions

- To account for both canonical and non-canonical data of CM-sluicing, especially observing their putative clausal source.
- To observe the distinct characteristics that CM-sluicing has through empirical data.

- For this study, COCA (Corpus of Contemporary American English) was used, from all of the nine registers that are available in the corpus.
- The search strings used are:
 - (11) a. * wh* and | or wh* PUNC
 - b. * how and | or wh* PUNC
 - c. * wh* and | or how PUNC
 - d. * which NOUN and | or wh* PUNC
 - e. * wh* and | or which NOUN PUNC
 - f. * which NOUN and | or how PUNC
 - g. * how and | or which NOUN PUNC
- From over 652 results, a total of 284 tokens were manually sorted for under discussion.

Remnant pairs

- The data consisted of a diverse set of *wh*-remnant pairs, which can be organized in argument-argument, argument-adjunct, adjunct-argument, and adjunct-adjunct pairs.
- (12)
- a. Someone was keeping something a secret. The only way to get Gella out of there is to find out **what** and **who**. (COCA 1991 FIC) *argument-argument pair*
 - b. He's moving something with all new guys. I don't want to tell you **what** or **where**. (COCA 2006 FIC) *argument-adjunct pair*
 - c. I knew I had to do something. I had no idea **why** or **what**. (COCA 1997 FIC) *adjunct-argument pair*
 - d. Breast cancer can be beaten, they say, and they are here to tell **how** and **why**. (COCA 2002 MAG) *adjunct-adjunct pair*

Remnant pairs distribution

- The table below shows the distribution of the remnant pairs.

	<i>and</i>	<i>or</i>
argument-argument	1	11
argument-adjunct	18	28
adjunct-argument	2	13
adjunct-adjunct	60	151
total	81	203

Table 1: Remnant pairs by conjunctions

- CM-slucing prefers the conjunction ‘or’ over ‘and’, and also the ‘adjunct-adjunct’ pair the most out of the four remnant pair types.
- Data with argument-denoting remnants make up 73 tokens of the data, which is about 25% of the 284 data collected.

- All 284 tokens of CM-sluicing had overt linguistic antecedent, while correlates such as ‘something, someone, etc.’ and other indexed NPs were either overt or covert.
 - (13) a. Someone was keeping something a secret. The only way to get Gella out of there is to find out **what** and **who**. (COCA 1991 FIC) *overt correlate*
 - b. We need to know more about measurement to decide whether to screen, and if so, **how** and **whom**. (COCA 1991 MOV) *covert correlate*
- The ratio from the raw frequency of CM-sluicing cases with arguments that had overt correlates (33 tokens, 45%) and covert correlates (40 tokens, 55%) were similar.

Antecedent clause sharing

- This variable observed the relationship between the reconstructed forms of the remnants and their linguistic antecedents, regarding the degree of likeness of their shared form.

(14) It was burned down in 1899, but no one seems to know **how** or **why**. (COCA 1996 FIC) *Completely sharing*

(15) a. Something had exploded, but none of the men and women could say **what** <had exploded> or **why** <**it** had exploded>. (COCA 2008 FIC) *Partial sharing with insertion*

b. I always knew that one day I would write and publish a novel, the question was never if but rather **what** <I would write> and **when** <I would write **it**>. (COCA 2012 TV)

Partial sharing with a predicate

Antecedent clause sharing distribution

- The table below shows that there are only two degrees regarding the antecedent and the putative source sharing, either completely or partially sharing.

<i>and or</i>		
completely	60	173
partially	21	30
total	81	203

Table 2: Antecedent source sharing by conjunctions

- CM-sluicing prefers to have their putative source of the remnants to be identical with the antecedent clause.
- Examples that partially share makes up 18% (51 tokens) of the total data, which requires an approach that can accommodate both of these instances.

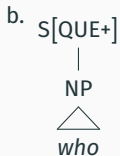
A Direct Interpretation (DI) approach

- **A Direct Interpretation (DI) approach** assumes that the meanings of the unpronounced material is generated without any syntactic structure. Instead, the sole daughter of an S-node, a fragment, is directly projected to its maximal projection (see Ginzburg and Sag 2000; Culicover and Jackendoff 2005).

(16) **Head-Fragment Construction** (Kim and Nykiel 2020):

Any category can be projected into a NSU (non-sentential utterance) when it functions as a salient utterance (SAL-UTT).

(17) a. Kim met someone, but I don't know who.



Dialogue and Question Under Discussion

- The interpretation of a sluiced clause depends on the notion of ‘question-under-discussion (QUD)’ in the dialogue, which are lists of possible answers to the current question (Ginzburg and Sag 2000).
- Dialogues are described via a Dialogue Game Board (DGB) where the contextual parameters are ‘anchored’ and where there is a record of who said what to whom, and what or who they were referring to (see Ginzburg 1996, Ginzburg and Fernandez 2010).

An example

- Uttering the question *Who did Kim meet?* will activate the following feature structure with the appropriate DGB information, where each *wh*-question is treated as a proposition in a question (Kim 2015):

(18) Semantic content of *who*: $\pi^i_{\{\text{person}(i)\}}$

(19)

$$\left[\text{FORM } \langle \text{Who did Kim meet?} \rangle \text{ SYN } S \text{ SEM } \lambda \left\{ \pi^i \right\} \left[\text{meet}(k, i) \right] \text{ DGB } \left[\text{MAX-QUD } \lambda \right] \right]$$

Argument-argument: Partial sharing

- There is one shared QUD with each *wh*-expression referring to a different argument variable.
- There must be two variables, because each remnant refers to a different entity, as can be inferred from distinct correlates of the antecedent.

- (20) a. Someone was keeping something a secret. The only way to get Gella out of there is to find out **what**_y and **who**_x. (COCA 1991 FIC)
- b. Structure: ... find out [_S [_{NP} what]] and [_S [_{NP} who]]
- c. QUD: $\lambda_x \lambda_y [\text{keep}(x, y, \text{secret})]$

Adjunct-adjunct pairs: Completely sharing

- There is one shared QUD but each sluiced construction is linked to a different variable (manner and reason).

(21) a. Breast cancer can be beaten, they say, and they are here to tell **how** and **why**.

(COCA 2002 MAG)

b. Structure: ...tell [_S [_{Adv} how]] and [_S [_{Adv} why]]

c. QUD: $\lambda_x \lambda_y [\text{tell}(\text{they}, \text{beat}(\text{we}, \text{cancer}, x[\text{how}], y[\text{why}])))]$

Argument-adjunct: Completely sharing

- There is only one shared QUD while each sluiced is linked to a different variable.
(22) a. He's moving something with all new guys. I don't want to tell you **what**_x
or **where**. (COCA 2006 FIC)
b. Structure: ..tell [_S [_{NP} what]] and [_S [_{Adv} where]]
c. QUD: $\lambda_x \lambda_y [-\text{tell}(I, \text{you}, \text{move}(h, x[\text{something}], \text{with.the.guys}, y[\text{where}])))]$

Adjunct-argument: Completely sharing

- There is only one shared QUD while each sluiced one refers to a different variable.
- Adjunct-argument pair can be analyzed in the same way as argument-adjunct cases in QUD, even though their ordering is different in SAL-UTT.

(23) a. I knew I had to do something. I had no idea **why** or **what**_x. (COCA 1997 FIC)

b. Structure: ... no idea [_S [_{Adv} why]] and [_S [_{NP} what]]

c. QUD: $\lambda_x \lambda_y [\neg \text{idea}(\text{do}(\text{I}, \mathbf{x[\text{something}]}, y[\text{why}])))]$

- The DI approach is able to resolve syntactic identity issues such as voice mismatches.

- (24) a. And that's all I'm saying, is if you start to look for the
why Kennedy was killed, if you start to look for a motive, then you can
begin to look for **who**_x and **how**. (COCA 1991 SPOK)
- b. Structure: ... look for [_S [_{NP} who]] and [_S [_{Adv} how]]
- c. QUD: $\lambda_x \lambda_y [\text{kill}(x, \text{Kennedy}, y[\text{how}])]$

- The QUD evoked here could be 'who killed Kennedy how', in which each sluiced remnants are each linked to a different variable.

- CM-sluicing is relatively flexible construction that is able to accommodate various structures as its antecedent.
- The distribution of CM-sluicing in COCA shows that it can be divided into two subtypes: examples that completely or partially sharing.
- Each clause of the CM-sluicing shared one QUD, while each sluiced phrase is linked to a different variable.
- The DI approach requires no complex syntactic structure for the semantic resolution. This appears to be much a simpler account for the CM-sluicing examples.

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